

Quantum GNN for Supplier Procurement Risk Analysis

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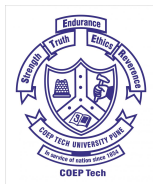
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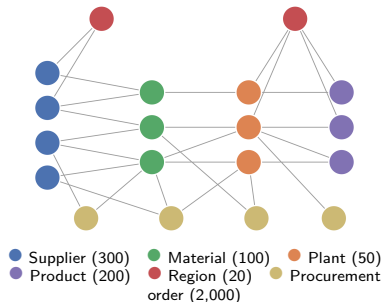
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Introduction

- **Procurement:** sourcing materials from external suppliers so production can continue
- A supply chain is a **network**; a problem at one node **propagates along it**
- **Risk assessment:** predict which suppliers are likely to be disrupted, and how severely
- Hard: disruptions are **rare** (class imbalance) and **severe** events differ from normal history



Illustrative subgraph; full graph: 2,670 nodes, 7,675 edges

- **GNN:** each node builds an embedding by aggregating its neighbours' information. **GraphSAGE** does this by sampling neighbours, so it scales
- For risk: suppliers are nodes, dependencies are edges, output is a **risk score per supplier**
- Classical limits: **over-smoothing** and **over-squashing**
- **QGNN:** a variational quantum circuit replaces part of the model. Thought to help (hypotheses we test):
 - Encoding into a **high-dimensional Hilbert space**
 - **Entanglement** models feature correlations compactly
 - Very few parameters, suited to small data and rare events
- **Our design:** a small VQC head on a **frozen GraphSAGE embedding**, as a controlled comparison with classical GraphSAGE — not a quantum-advantage claim

Literature Survey

Table 1: Literature Survey

Ref	Author (Year)	Focus / Method	Limitation
[1]	Fareed (2026)	Classical ML for supplier risk prediction and resilience scoring	Preprint, not peer-reviewed; purely classical, no quantum baseline
[2]	Ziaria et al. (2026)	ML-based supplier evaluation; sustainable supplier selection and delay-risk framework	Classical ML only; no graph structure or quantum component
[3]	Rahman & Nguyen (2026)	QCNN survey; entanglement-based correlation learning	Survey only; no supply-chain or graph evaluation
[4]	Lamichhane & Rawat (2025)	Survey of QML: advances, challenges, perspectives	Survey only; no empirical evaluation on any domain task
[5]	Larbi et al. (2025)	Dual ML models for IoT component failure and parts demand	Classical ML only; no quantum formulation

Literature Survey

Table 1: Literature Survey (Contd.)

Ref	Author (Year)	Focus / Method	Limitation
[6]	Innan et al. (2024)	QGNN with VQC for credit-card fraud detection	Simulator only; 6 qubits; heavy undersampling; $\approx 3\%$ gain over classical GNN
[7]	Tudisco et al. (2024)	VQC classifier on real bank transaction data	Dataset reduced in size and features via PCA due to NISQ limits; scalability unproven
[8]	Soni et al. (2024)	Survey of quantum optimization in supply chain management and logistics	Survey only; no learned risk-prediction model
[9]	Jiang, Shen & Liu (2022)	Quantum inventory model; supply-chain structures suited to quantum methods	Inventory setting; no learned risk prediction or GNN
[10]	Yadav (2021)	Survey of GCN/GAT risk propagation; GNNs model multi-hop supply-chain dependencies	Survey only; classical GNNs, no quantum formulation

Motivation and Need for Study

- **Same structural problem as fraud:** rare, high-impact events in a relational network with extreme class imbalance — the setting where QGNN and VQC work has been reported [6, 7]
- **Supplier networks are natively graphs:** suppliers, hubs and raw-material sources as nodes; dependency and material flow as edges
- **Classical GNNs struggle where procurement risk lives:** over-smoothing flattens embeddings at depth, over-squashing bottlenecks multi-hop signal
- **Gap in the surveyed work:**
 - Supplier-risk studies are purely classical [1, 2, 5]
 - Quantum studies target finance, are surveys, or address inventory [3, 4, 6–9]
 - No QGNN applied to supplier procurement risk
- **Need:** a **controlled comparison** against a classical GraphSAGE baseline, so the outcome is defensible either way

Problem Statement

To investigate whether a **hybrid quantum-classical head**, a small variational quantum circuit reading a frozen GraphSAGE supplier embedding, can match or exceed a **classical GraphSAGE** baseline for **supplier disruption-risk prediction**, and to identify which architectural and training factors govern its behavior.

- Evaluated on temporal and severity (out-of-distribution) splits of a synthetic supply-chain benchmark

Research Objectives

- 1 **Formulate** supplier procurement risk as a node-level prediction task on a supply-chain graph
- 2 **Implement** a classical GraphSAGE baseline for supplier disruption-risk prediction
- 3 **Implement** a hybrid QGNN: a variational quantum circuit head on the frozen GraphSAGE embedding
- 4 **Compare** the classical and quantum models under identical data, splits and metrics
- 5 **Analyze** which design factors (encoding, qubits, depth) affect the hybrid model's behavior

Software Requirements Specification

Table 2: Functional and Non-Functional Requirements

Requirement	Description
<i>Functional</i>	
Data ingestion and graph construction	Load the synthetic supplier dataset and build a heterogeneous supply-chain graph (6 node types, 9 relation types)
Feature encoding	Encode static and time-varying supplier attributes as node features
Hybrid risk model	GraphSAGE encoder with a VQC head producing a supplier disruption-risk score
Disruption simulation	Black-swan events simulated over 104 periods (about 2 years, weekly)
Evaluation and reporting	PR-AUC and related metrics on temporal and severity splits, with tables and figures
<i>Non-functional</i>	
Fair comparison	Classical and quantum models share the same data, splits and metrics
Reproducibility	Fixed random seeds and saved experiment artifacts
Scalability	Pipeline designed for larger graphs (currently 2,670 nodes, 7,675 edges)

SRS Contd...: Technology Stack

Table 3: Technology Stack

Category	Technology / Tool
Programming language	Python 3.12
Deep learning	PyTorch
Graph learning	PyTorch Geometric (HeteroConv, SAGEConv)
Quantum framework	PennyLane (default.qubit, ideal noise-free simulator)
Graph analysis	NetworkX (topology validation)
Classical ML diagnostics	scikit-learn
Dataset	Synthetic multi-tier supply-chain graph benchmark: 2,670 nodes, 7,675 edges, 104 periods
Hardware	CPU only; no physical quantum hardware used

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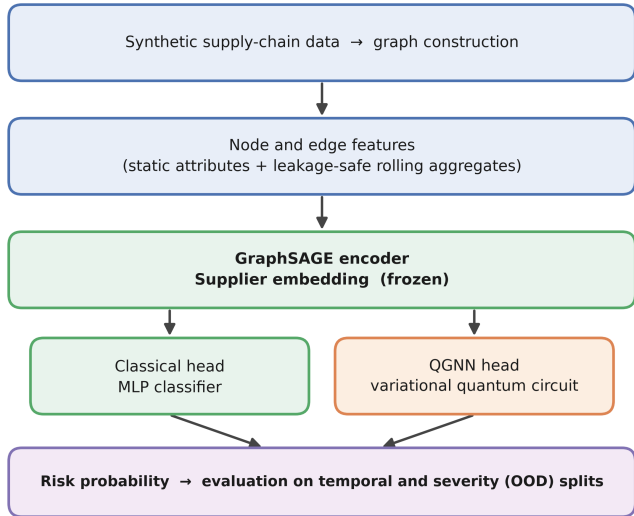
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SRS Contd...: System Diagram



Concrete Future Plan

Table 4: Concrete Future Plan

No.	Stage	Planned Work (one factor varied at a time)
1	Benchmark and graph	Synthetic supply-chain dataset with black-swan events; heterogeneous graph construction
2	Classical baseline	GraphSAGE encoder trained and evaluated on temporal and severity splits
3	QGNN baseline	VQC head on the frozen GraphSAGE embedding; training stabilisation (patience, output scaling, LayerNorm)
4	Circuit ablations	Qubit count (4, 6, 8) and circuit depth (1, 2, 3 layers)
5	Representation audit	Diagnostic classifiers on the frozen embedding to check its signal
6	Head design	Projection architecture and parameter initialization
7	Ansatz and encoding	Entanglement topology; encoding scale, type and data re-uploading
8	Final evaluation	Classical vs. QGNN over multiple seeds: ranking, calibration and stability metrics

Expected Outcomes

Table 5: Expected Outcomes

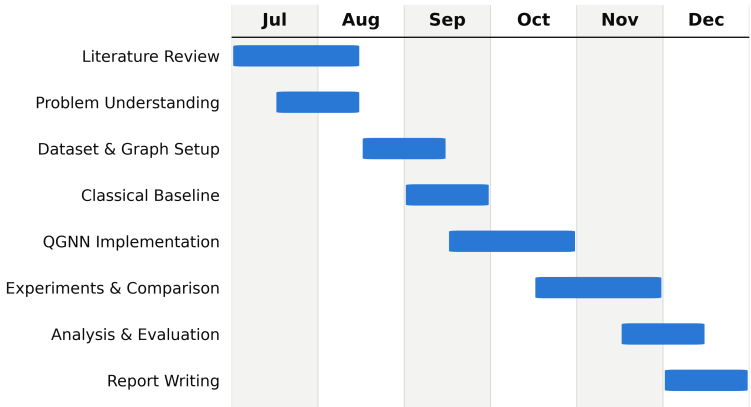
No.	Expected Outcome	How it will be assessed
1	Working classical GraphSAGE baseline and hybrid QGNN for supplier disruption-risk prediction	Both implemented on one shared data and training pipeline
2	A controlled comparison of the two models	Same data, splits, metrics and multiple seeds; PR-AUC as primary metric
3	Understanding of how each model handles normal vs. severe conditions	Temporal split and severity (out-of-distribution) split, reported separately
4	A map of which design factors influence QGNN behavior	One factor varied at a time: qubits, depth, ansatz, encoding, initialization
5	Insight into how much risk signal the frozen GraphSAGE embedding carries	Diagnostic classifiers on the frozen embedding, compared with both model heads
6	Clearly scoped conclusions	Stated for a synthetic benchmark on an ideal simulator, not as a general quantum-advantage claim

Summary

- **Problem:** supplier procurement risk is a rare-event prediction task on a network where disruptions propagate
- **Gap:** supplier-risk studies are classical, and quantum studies target other domains such as finance
- **Idea:** pair a classical GraphSAGE encoder with a small variational quantum circuit head for supplier risk prediction
- **Approach:** a controlled comparison against a classical GraphSAGE baseline, studying which design factors matter
- **Expected contribution:** an evidence-based, clearly scoped understanding of when a quantum head helps, not a claim of quantum advantage

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Questions/ Discussion